IoT-Based Smart Soil Health Monitoring and Crop Recommendation System

A Project Work Synopsis

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Abstract

Agriculture acts as the primary economic backbone for many developing nations, yet the sector faces significant challenges due to inefficient resource management and a lack of technological integration. Traditional farming methods heavily rely on manual soil testing, which is often time-consuming, infrequent, and prone to human error. This lack of real-time data leads to the excessive application of fertilizers, soil degradation, and suboptimal crop selection.

This project proposes the design and implementation of an IoT-Based Smart Soil Health Monitoring and Crop Recommendation System. The system utilizes a network of sensors to monitor critical soil parameters specifically Nitrogen, Phosphorus, and Potassium (NPK), soil moisture, temperature, and humidity in real-time. The collected data is processed by a microcontroller and transmitted to a cloud platform for analysis. A key innovation of this system is its intelligent decision-making engine, which compares the live soil data against a standard agricultural database to generate actionable crop recommendations and precise fertilizer dosage requirements. By automating the soil analysis process, this solution empowers farmers to make data-driven decisions, thereby optimizing input costs, improving crop yields, and promoting sustainable precision agriculture.

Keywords:

1. Internet of Things (IoT)
2. Smart Agriculture
3. Soil Health Monitoring
4. NPK Sensors
5. Crop Recommendation System
6. Precision Farming
7. Data Analytics

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1. INTRODUCTION

Agriculture is the backbone of the economy for many developing nations, yet the sector faces significant challenges due to inefficient resource management and a lack of technological integration. Soil health is the most critical factor influencing crop productivity; however, traditional farming methods often rely on intuition or historical practices rather than precise data. This leads to the overuse of chemical fertilizers, soil degradation, and inconsistent yields.

To address these issues, the integration of the Internet of Things (IoT) in agriculture often termed "Smart Farming" has emerged as a viable solution. By utilizing sensors and automated decision-making systems, farmers can monitor field conditions in real-time and optimize their resource usage.

1.1 Problem Definition

Farmers currently lack access to affordable, real-time, and automated soil analysis systems. Conventional soil testing methods involve sending samples to laboratories, a process that is time-consuming (often taking weeks) and logistically difficult for remote farmers. As a result, critical decisions regarding crop selection and fertilizer application are often made without accurate data, leading to economic loss and environmental damage.

1.2 Problem Overview

This project proposes the design and development of an IoT-Based Smart Soil Health Monitoring and Crop Recommendation System. The system utilizes a network of sensors to monitor essential soil parameters specifically Nitrogen, Phosphorus, and Potassium (NPK), moisture, temperature, and humidity in real-time. The collected data is processed by a microcontroller and transmitted to a cloud platform. A key feature of this project is its ability to not just display data, but to analyse it and provide actionable recommendations for suitable crops and precise fertilizer dosages.

1.3 Hardware Specification

- **Microcontroller:** ESP32 (Wi-Fi enabled) or Arduino Mega with Wi-Fi Module.
- **Soil Nutrient Sensor:** Modbus RS485 NPK Sensor (for detecting N, P, K levels in mg/kg).
- **Soil Moisture Sensor:** Capacitive type (corrosion resistant).

- **Environmental Sensor:** DHT11 or DHT22 (for Temperature and Humidity).
- **Display:** 0.96-inch OLED Display (for on-site readings).
- **Power Supply:** 12V DC Adapter or Li-Ion Battery Pack.

1.4 Software Specification

- **Firmware Development:** Arduino IDE (C++).
- **Cloud Platform:** Thing Speak, Blynk, or Firebase (for real-time data logging).
- **Data Analysis:** Python script (for recommendation logic).
- **User Interface:** Mobile Application (Blynk/Custom) or Web Dashboard.

2. LITERATURE SURVEY

2.1 Existing System

Currently, the most common method for soil analysis is laboratory testing. Farmers collect soil samples from various parts of their field and transport them to a government or private lab.

- **Disadvantages:** High latency (delay in results), one-time snapshot (not continuous monitoring), and potential for human error during sampling or transport.

2.2 Proposed System

The proposed system replaces manual sampling with in-situ sensors that provide continuous data.

- **Advantages:** Real-time results, historical data tracking, automated alerts, and logic-based recommendations that do not require an agricultural expert to interpret.

2.3 Literature Review Summary

Year and Citation	Article/ Author	Tools/ Software	Technique	Source	Evaluation Parameter
2017 [1]	Prathibha et al.	Temp Sensors	Zigbee	IEEE Xplore (ICRAECT)	**Finding:** successfully automated irrigation based on moisture. **Gap:** Did not monitor soil nutrients (NPK)
2018 [2]	Vineela et al.	Arduino, GSM Module	SMS Alert System using GSM	IJERT Journal	Alert Latency & User Notification Speed
2019 [3]	Raj et al.	Raspberry Pi, Camera Module	Image Processing (Colorimetry)	IJARCS Journal	NPK Detection Accuracy via Image
2020 [4]	Girish et al.	IoT Cloud, Humidity Sensors	Cloud based Data. Logging	EEE Xplore (CONECCT)	Data Transmission Reliability
2021 [5]	Sowmya et al.	Python, Scikit-learn	Machine Learning (Random Forest)	IEEE Xplore (ICAIS)	Prediction Accuracy of Soil Properties

3. PROBLEM FORMULATION

The project aims to solve the following core problems:

1. Latency in Decision Making: Eliminating the waiting period associated with lab testing.

2. Blind Resource Usage: preventing the overuse of fertilizers which increases cost and damages soil biodiversity.
3. Crop Mismatch: Reducing instances where farmers plant crops unsuited to the current soil fertility profile.

Formal Statement: "*To design and implement a low-cost, real-time IoT system that accurately measures soil NPK and environmental parameters and utilizes this data to generate algorithmic recommendations for crop selection and fertilizer application.*"

4. OBJECTIVES

The project aims to solve the following core problems:

1. **Latency in Decision Making:** Eliminating the waiting period associated with lab testing.

2. **Blind Resource Usage:** preventing the overuse of fertilizers which increases cost and damages soil biodiversity.

3. **Crop Mismatch:** Reducing instances where farmers plant crops unsuited to the current soil fertility profile.

Formal Statement: *To design and implement a low-cost, real-time IoT system that accurately measures soil NPK and environmental parameters and utilizes this data to generate algorithmic recommendations for crop selection and fertilizer application.*

5. METHODOLOGY

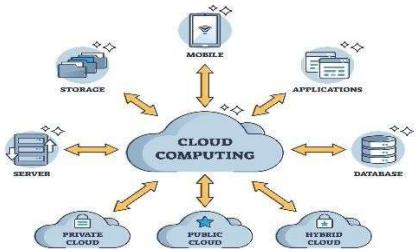
The project will be implemented in the following phases:

Phase 1: Data Acquisition (Perception Layer)

- Sensors (NPK, Moisture, DHT11) will be deployed in the soil.
- The sensors will convert physical parameters into analogue/digital signals.

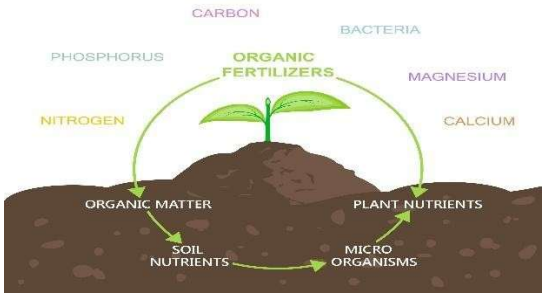
Phase 2: Data Processing (Network Layer)

- The ESP32 microcontroller will read the sensor data.
- Data will be formatted into JSON packets.
- The ESP32 will transmit this data over Wi-Fi to the Cloud Server.



Phase 3: Data Analytics (Application Layer)

- The Cloud platform (e.g., Thing Speak) will store the data.
- A backend script (Python/MATLAB) will fetch the data and compare it against a "Crop Database" (e.g., if N > 50, recommend Wheat).



Phase 4: User Interaction

- The final output (Crop Name + Fertilizer Quantity) will be displayed on the farmer's mobile app.

6. EXPERIMENTAL SETUP

Hardware Setup: The prototype will be assembled on a PCB or Breadboard. The NPK sensor (Modbus) will be connected via a MAX485 TTL to RS485 converter module to the ESP32. The system will be powered by a 12V supply to ensure accurate readings from the industrial-grade NPK sensor.

Testing Plan

1. **Unit Testing:** Each sensor will be tested individually for data consistency.
2. **Integration Testing:** The complete system will be tested to ensure data is successfully uploaded to the cloud without packet loss.
3. **Field Validation:** The device will be tested in three different soil types (Sandy, Loamy, Clay) to verify the robustness of the recommendation logic.

7. CONCLUSION

The IoT-Based Smart Soil Health Monitoring and Crop Recommendation System aims to modernize traditional farming. By providing farmers with immediate insights into their soil's health, the system empowers them to make data-driven decisions. This leads to optimized input costs (less fertilizer), improved crop yields, and sustainable agricultural practices. The project serves as a significant step towards Precision Agriculture.

8. REFERENCES

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